

2.20 NATURAL VENTILATION

ADG Development: refer to Part 4B of the ADG.

Objective

- (a) To ensure dwellings are designed to provide all habitable rooms with direct access to fresh air and to assist in promoting thermal comfort for occupants.
- (b) To encourage the design of the development to address orientation, building envelope and the internal configuration of dwellings.
- (c) To provide natural ventilation in non-habitable rooms, where possible.
- (d) To reduce the use of mechanical ventilation, particularly air-conditioning.

Controls

- (a) All dwellings in a development are to be naturally cross-ventilated. These may be either dual aspect (e.g. cross through dwellings and corner dwellings), or maisonette/2 storey dwellings which draw cool air in at lower levels and allow warm air to escape at higher levels.
- (b) Plan the site to utilise natural breezes by:
 - (i) Determining prevailing breezes and orienting buildings to maximise access to breezes, where possible;
 - (ii) Locating vegetation to direct breezes and cool air as it flows across the site; and
 - (iii) Selecting and planting trees that do not inhibit airflow.
- (c) Design the internal dwelling layout to promote natural ventilation by minimising interruptions (such as corners and walls) to air flow through a dwelling.
- (d) Doors and operable windows are to maximise natural ventilation by:
 - (i) Locating small windows on the windward side and larger windows on the leeward side of the building, allowing air pressure to draw air through the dwelling;
 - (ii) Using higher level casement or sash windows, clerestory windows or operable fanlight windows to facilitate convective currents; and
 - (iii) Selecting windows which can be reconfigured to funnel breezes into the dwelling.
- (e) Innovative technologies to naturally ventilate internal rooms such as laundries, bathrooms and basement car parks are to be implemented including stack-effect ventilation or solar chimneys.
- (f) To minimise use of air-conditioning, all dwellings must have ceiling fans installed in all habitable rooms.